Basics of Computer Networks — Course Plan

Cyber.SoHo Educational Hub • *From the Industry to the Classroom — Building the IT's Future, Today and Together!*

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1. Course Identification

Field	Value
Course Title	Basics of Computer Networks
Status	Non-credited (no exams, no homework grading, no evaluation)
Audience	College students (introductory, no prerequisites)
Total Hours	24 contact hours
Schedule	6 days × 4 contact hours/day (2 lectures + 2 labs/day)
Term	April 28, 2026 → May 14, 2026
Calendar	Tue / Thu, 6 sessions over 17 calendar days
Ponderation	**2 – 2 – 2** (2 h lecture, 2 h lab, 2 h home practice — per session)
Format	Hybrid-friendly (in-class + virtualized lab)
Reference Curriculum	Cisco NetAcad — Networking Essentials 3.0

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2. Course Description

This is a hands-on, **project-based introduction** to computer networking. Across six 4-hour sessions, students plan, build, address, secure, and troubleshoot a complete **Small Office / Home Office (SoHo)** network using free virtualization & simulation tools. By the end, students will have a working **GNS3** lab on their own machine, a documented topology, and the foundational knowledge to pursue Cisco NetAcad's **Networking Essentials 3.0** or **CCNA — Introduction to Networks (ITN)** as a next step.

The course is non-credited: there are **no quizzes, no exams, no homework submissions, and no grades**. All learning is reinforced by the running 6-day project (the *Cyber.SoHo Network Build Challenge*) and weekly check-ins.

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3. Learning Outcomes

By the end of the course, students will be able to:

1. Explain core concepts of network communication — data, speed/capacity, the roles of clients, servers and infrastructure devices.
2. Identify common cabling types (twisted-pair, coax, fiber-optic) and Ethernet signaling.
3. Compare the **OSI** (Open Systems Interconnection) and **TCP/IP** (Transmission Control Protocol / Internet Protocol) reference models, including encapsulation and Ethernet framing.
4. Build a simulated **LAN** (Local Area Network) using GNS3, Wireshark, and either VirtualBox or KVM.
5. Apply IPv4 addressing — convert binary↔decimal, identify classes, separate public/private ranges, and distinguish unicast / multicast / broadcast.
6. Configure **DHCP** (Dynamic Host Configuration Protocol) to dynamically assign IPv4 addresses; understand **NAT** (Network Address Translation) and the rationale for **IPv6**.
7. Describe and demonstrate common application services: **ARP**, **DNS** (Domain Name System), **HTTP** (Hypertext Transfer Protocol), **FTP** (File Transfer Protocol), **SSH** (Secure Shell), and email protocols.
8. Configure a basic integrated wireless router and explain how Wi-Fi traffic is controlled.
9. Recognize common security threats (malware, social engineering, **DoS** — Denial of Service, brute force) and apply elementary mitigations.
10. Use **Nmap** / **Zenmap** ethically to inventory their own lab; use Cisco-style basic commands to verify connectivity and troubleshoot.

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4. Lab Environment — Free Tools (All Working Links)

Students may pick **any** of the following hypervisors / simulators for the labs. The course is platform-neutral, but the official walkthroughs use **Oracle VirtualBox + GNS3 + Wireshark** as the reference stack (free, cross-platform, no registration required for the core).

Hypervisors / Type-1 & Type-2

Tool	Notes	Link
Oracle VirtualBox	Free, cross-platform, GPL-v3. **Reference for this course.**	https://www.virtualbox.org/
KVM (Kernel-based Virtual Machine)	Linux native, type-1 hypervisor.	https://www.linux-kvm.org/page/Main_Page
Proxmox VE (Virtual Environment)	Open-source server hypervisor + web UI.	https://www.proxmox.com/en/proxmox-virtual-environment/overview
VMware Workstation Pro	Free for personal, educational &	https://www.vmware.com/products/desktop-hypervisor/workstation-and-fusion

Tool	Notes	Link
	commercial use since Nov 2024 (Broadcom).	

Network Simulators / Emulators

Tool	Notes	Link
GNS3 (Graphical Network Simulator-3)	**Reference for this course.** Real Cisco IOS images + integration with VirtualBox/KVM/VMware.	https://www.gns3.com/
Cisco CML — Free Tier (Cisco Modeling Labs)	5-node free tier for personal use. Requires free Cisco account.	https://learningnetwork.cisco.com/s/cml-personal
Cisco Packet Tracer	Free via NetAcad — best for early learners and Cisco-specific labs.	https://www.netacad.com/cisco-packet-tracer
Wireshark	Industry-standard packet analyzer. **Required.**	https://www.wireshark.org/

⚠️ **VirusTotal Reminder** — Before installing any tool, students should verify the installer's hash on https://www.virustotal.com/. Always download from the **official** vendor site listed above.

Minimum Host Requirements

- **64-bit OS** — Windows 10/11, macOS, or any modern Linux distro
- **CPU** — VT-x / AMD-V enabled in firmware
- **RAM** — 8 GB minimum (16 GB strongly recommended for GNS3 + VMs)
- **Disk** — 30 GB free
- **Network** — Reliable broadband for downloading appliance images

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5. Pedagogy — Project-Based Learning

The full **24 hours of class time + 12 hours of suggested home practice (12h)** are organized around **one continuous project** that runs from Day 1 to Day 6:

🛠️ The *Cyber.SoHo Network Build Challenge*

You are the **junior network admin** for a fictional 8-person consulting firm — *Acme SoHo Consulting* — moving into a new two-floor office. By the end of Day 6 you must hand the senior

admin a fully simulated, addressed, secured, and documented network running entirely inside GNS3 on your laptop.

Each session contributes one **project stage** (E2 lab + 2 h home practice). The lectures and E1 labs each session deliver the concepts and tools you'll apply to that stage. **No grading, but stages stack** — falling behind a stage hurts the next stage.

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6. Detailed 6-Day Schedule

Legend • **L1, L2** = lectures (1 h each) • **E1, E2** = exercises / labs (1 h each) • 🏠 = home practice (≈ 2 h, optional but recommended)

📅 Day 1 — Tuesday April 28, 2026 — *Foundations: How Networks Talk*

Block	Topic
L1	What is a network? Network data. Transmission speed & capacity (bps, kbps, Mbps, Gbps). Roles of **clients**, **servers**, and infrastructure devices (hubs, switches, routers, **APs** — Access Points, firewalls). Types of networks: **PAN** (Personal Area Network), **LAN**, **MAN** (Metropolitan Area Network), **WAN** (Wide Area Network).
E1	**Lab tools tour** — verify CPU virtualization is enabled in firmware; install VirtualBox; install GNS3 + GNS3 VM; install Wireshark; smoke-test by capturing your own loopback traffic.
L2	Getting online: requirements for host connectivity (IP address, subnet mask, default gateway, DNS). Mobile networks (3G / 4G / 5G / Wi-Fi). The importance of **network documentation** (logical vs physical diagrams, device inventory, IP plan, change log).
E2	🚀 **Project Stage 1 — Discover & Plan.** Draft the *Acme SoHo* requirements document (1 page) and a hand-drawn / drawio topology sketch: 8 endpoints, 1 Wi-Fi segment, 1 server, 1 internet uplink.
🏠	Read NetAcad NetEss 3.0 *Module 1 — Communications in a Connected World* and *Module 2 — Network Components, Types, and Connections*. Refine the topology sketch based on what you learn.

📅 Day 2 — Thursday April 30, 2026 — *Cabling, Standards & Reference Models*

Block	Topic
L1	Common network cables: copper twisted-pair (**UTP** — Unshielded Twisted-Pair, **STP** — Shielded Twisted-Pair), coaxial, **fiber-optic** (single-mode vs multi-mode). How a

Block	Topic
	twisted-pair cable sends and receives signals — differential signaling, attenuation, **EMI** (Electromagnetic Interference). Common categories: **Cat 5e**, **Cat 6**, **Cat 6a**, **Cat 8**.
E1	**GNS3 deep dive** — touring the user interface, projects, templates, the GNS3 Marketplace, importing **VMs** (Virtual Machines) from VirtualBox / VMware; basic console & link operations.
L2	The role of standards (**IEEE** — Institute of Electrical and Electronics Engineers, **IETF** — Internet Engineering Task Force, **ISO** — International Organization for Standardization). Comparing the **OSI 7-layer model** with the **TCP/IP 4-layer model**. Explain Layer 1 (Physical) and Layer 2 (Data Link) in an Ethernet network.
E2	🚀 **Project Stage 2 — Build the topology.** In GNS3 build: 1 Cisco router, 1 multilayer switch, 2 endpoints (VPCS or VirtualBox VM). Verify L1 link-up.
🏠	Read NetAcad NetEss 3.0 *Module 6 — Network Media* and *Module 5 — Communications Principles*. Practice 5 OSI vs TCP/IP mapping exercises.

📅 Day 3 — Tuesday May 5, 2026 — *Ethernet & Routing*

Block	Topic
L1	How Ethernet communicates: **MAC** (Media Access Control) addresses, the Ethernet frame, encapsulation across the OSI stack, broadcast domains. The **3-layer hierarchical design model** (access, distribution, core). Why we contain broadcasts.
E1	**Wireshark lab** — capture an Ethernet frame, decode source/destination **MAC**, **EtherType**, payload; identify ARP and ICMP.
L2	The need for routing. How routers use routing tables (directly connected, static, dynamic). Building a fully connected network — what changes when you cross a Layer-3 boundary.
E2	🚀 **Project Stage 3 — Wire the LAN.** Add a second **LAN** segment to your project. Configure router interfaces. Verify ping between segments.
🏠	NetAcad NetEss 3.0 *Module 21 — Ethernet Switching* + *Module 22 — Network Layer*. Repeat the Wireshark capture against a different protocol (e.g. **DHCP** — Dynamic Host Configuration Protocol).

📅 Day 4 — Thursday May 7, 2026 — *IP Addressing & Address Management*

Block	Topic
L1	The IPv4 address — 32-bit, dotted-decimal. Binary ↔ decimal conversion. Subnet masks. **Classes** (A, B, C, D, E) — historical view. Public vs private ranges (RFC 1918). Unicast, multicast, broadcast.
E1	**Subnetting workshop** — building a personal cheat-sheet (powers of 2, /24 → /30, magic-number method). 10 paper exercises.

Block	Topic
L2	Static vs dynamic addressing. **DHCP** — DORA (Discover, Offer, Request, Acknowledge). **NAT** — Network Address Translation (basic, **PAT** — Port Address Translation). Why **IPv6** will replace IPv4 (address exhaustion, simpler header). IPv6 features overview.
E2	🚀 **Project Stage 4 — Address the network.** Create your IP plan (subnet table). Configure DHCP on the router for one **LAN**, configure static IPs for the server VLAN.
🏠	NetAcad NetEss 3.0 *Modules 8-11* (Internet Protocol, IPv4 Segmentation, IPv6, DHCP). Subnet 192.168.50.0/24 into 4 equal subnets and document.

📅 Day 5 — Tuesday May 12, 2026 — *Application Services & Wireless*

Block	Topic
L1	Client-server interaction. **TCP** vs **UDP** (User Datagram Protocol) at the transport layer. Well-known port numbers (HTTP 80, HTTPS 443, SSH 22, FTP 21, DNS 53). Why **LAN segmentation** and **VLAN** (Virtual LAN) matter. Application-layer services: **ARP**, **DNS**, **HTTP/HTML**, **FTP**, **Telnet** vs **SSH**, email protocols (**SMTP** — Simple Mail Transfer Protocol, **POP3** — Post Office Protocol v3, **IMAP** — Internet Message Access Protocol).
E1	**Application services lab** — capture and decode an **ARP** request/reply, a **DNS** query, an **HTTP** GET in Wireshark.
L2	The integrated wireless router. Wired vs wireless. Wi-Fi standards (**802.11 a/b/g/n/ac/ax**). How wireless traffic is controlled (**CSMA/CA** — Carrier Sense Multiple Access with Collision Avoidance, channels, **SSID** — Service Set Identifier, beacons). Optional: configuring secure Wi-Fi (**WPA3** — Wi-Fi Protected Access 3, **PSK** — Pre-Shared Key).
E2	🚀 **Project Stage 5 — Add services & wireless.** Add a **DNS** + **HTTP** internal server. Add a Wi-Fi access segment (Cloud node + virtual AP, or simulated). Verify clients reach the web service by name.
🏠	NetAcad NetEss 3.0 *Modules 13-17* (ARP, DHCP, TCP/UDP, Application Services, Network Testing). Build the *Acme SoHo* DNS records list.

📅 Day 6 — Thursday May 14, 2026 — *Security & Troubleshooting (Final Stage)*

Block	Topic
L1	The **CIA triad** — Confidentiality, Integrity, Availability. Threat catalog: **social engineering** (phishing, vishing, smishing), malicious software (virus, worm, trojan, ransomware), **DoS** / **DDoS** (Distributed Denial of Service), brute force, password spraying. Antimalware basics. **Nmap** / **Zenmap** as host-discovery and service-enumeration tools — and the **ethics** of using them only on systems you own.

Block	Topic
E1	**Nmap / Zenmap lab** — scan **only your own GNS3 topology**. Identify hosts, open ports, services. Cross-reference findings to your IP plan.
L2	Configuring basic network security — secure user authentication, hardening Wi-Fi, basic firewall configuration on Cisco devices, Linux (iptables / nftables / ufw), Windows Server. The **troubleshooting methodology**: gather → isolate → resolve → document. Common Internet connectivity issues.
E2	🚀 **Project Stage 6 — Final integration & demo.** Add basic firewall rules to the router (allow established, deny unsolicited inbound from "internet" cloud). Walk a peer through your topology. Submit (informally) the final topology + IP plan + 1-page network documentation.
🏠	Read NetAcad NetEss 3.0 *Module 36 — Troubleshoot Network Issues*. Browse the Cisco NetAcad catalog and pick your next course (CCNA-ITN suggested).

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7. Required Materials & Costs

Item	Cost
Cyber.SoHo course materials (markdown lecture, slide deck, HTML module)	**$0**
Cisco NetAcad — Networking Essentials 3.0 (self-paced companion)	**$0** (sign up free)
Oracle VirtualBox + GNS3 + Wireshark + Cisco Packet Tracer	**$0**
Hardware: laptop with **8 GB RAM minimum** (16 GB recommended)	Student-supplied

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8. Teaching Methods

- **Live mini-lectures** with concrete real-world examples (industry-to-classroom).
- **Hands-on labs** in every session (50% of contact time).
- **Peer-paired project work** — students may work in pairs but each builds their own GNS3 file.
- **Ask-anything time** at the end of every session.

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9. Evaluation

None. This course is **non-credited**. There are no quizzes, no exams, no graded homework. The 6-stage project is the **only formal deliverable**, and it is reviewed informally for feedback only — not for a mark.

Students who want recognition can:

- Complete the free Cisco **NetAcad Networking Essentials 3.0** companion course online and earn the **Cisco-issued digital badge**.
- Take the optional **NetEss 3.0 final** on NetAcad after the course ends.

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10. Classroom Policies

- Be on time. Sessions are tight — 4 hours, 6 days.
- Be respectful — diverse backgrounds, single shared lab.
- **Lab ethics** — Nmap, Wireshark, and any reconnaissance tool are used **only** against your own GNS3 topology, **never** against the school network or any third-party system. This is a hard rule.
- Bring your laptop charged with VirtualBox + GNS3 pre-installed by Day 2.

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11. Disclaimers & Notes

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